

NAME: _____

PERIOD: _____

Which Design Supports Cause?

AP Statistics · Unit 1 · Topic 1.13

[STAR] THE PROBLEM

Suppose a researcher recruits 30 student volunteers to study whether an energy drink changes reaction time.

Design 1: The first 15 students to sign up receive the energy drink; the last 15 receive water. A tester who knows each drink measures reaction time in milliseconds.

Design 2: A coin flip assigns each volunteer to the energy drink or water. The water is flavored to match, both drinks are served in identical opaque cups, and neither the volunteer nor the tester knows which drink was assigned.

Two energy-drink designs

#	Drink allocation	Tester told drink?
1	First/last signup	Yes
2	Coin flip	No

FIRST TAKE — one sentence before you work the parts

Which design would make the comparison between drinks fairer? Give one reason.

[BOOK] PRINCIPLES OF EXPERIMENTAL DESIGN (keep on the page)

A well-designed experiment uses **comparison**, **random assignment**, **replication**, and control of potential confounding variables. A **control group** supplies a baseline. Random assignment tends to balance uncontrolled variables so a difference can be attributed to treatment. **Blinding** reduces changes caused by expectations or by how responses are measured. Selection still limits generalization.

Work (a)–(c).

DOK 1 (a) Name the experimental units and the response variable, with units.

DOK 2 (b) Compare how treatments are assigned in the two designs. Explain how one other feature of Design 2 helps the comparison.

DOK 3 (c) Which design gives stronger evidence about whether the drink causes a change? Explain one other reason the groups in Design 1 might react differently. State why the result would not automatically describe all teenagers.

Frame: Design _____ supports cause because _____.

Frame: The result cannot automatically generalize to _____ because _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.